

POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

An Approximation of the Error-Collection Methods of the Philippine Statistics Authority - Central Office

Presented to Prof. Lydinar Dastas

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The group extends their deepest appreciation for Information Management Prof. Lydinar Dastas for her unwavering insight and guidance to the group throughout the entire research and creation process and giving essential insight that helped the group achieve their goals in completing and testing the systems and research.

It is also with deep gratitude that the group acknowledges Sr. Alex M. Meriño for his immense assistance throughout the approval process for this project and for sharing his knowledge in regards to the Philippine Statistics Authority-Central Office and the workflow to give the group more information in regards to plotting and arranging the project to reflect the proper flow of the PSA.

INTRODUCTION

Overview of the Project

The Philippine Statistics Authority (PSA) is responsible for collecting and databaseing the many legal certificates pertaining to life in the Philippines. This includes Birth, Death, and Marriage Certificates which are stored for future reference and can be requested by the owner of the particular document. However, with the volume of data that the Philippines Statistics Authority (PSA) collects from other government sectors such as LGUs and Registry offices, many errors within the collected certificates appear. This is why the Philippines Statistic Authority has a system where erroneous data from collected forms and certificates from across the Philippines, collects them while also marking erroneous information based on cross references and requests before being entered into the system. However, it should be noted that this only affects the systems of the Philippine Statistics Authority and not the actual legal document submitted by the Local Civil Registry Office, or any other governing body as it is a different process independent of the PSA and their systems.

This problem has led the group to focus on the error collection system of the Philippine Statistics Authority as a basis for research in database management concepts and application as it is a manageable and database oriented topic which may be explored through simulation and study. Through the course of four months, the group has simulated a user-end data collection system through the creation of what the group approximates as a streamlined system designed to make erroneous data collection convenient and manageable both from the employee and administration side. The system incorporates SQL queries, Backend, and UI/IX elements to produce a fully-functioning demonstration of the system to catch any erroneous information present within This project was formulated with the guidance of Prof. Lydinar Dastas and coordination with the Philippine Statistics Authority as this project surrounds sensitive and personal information pertaining to systems and forms which belong to people or organizations, thus, all given details in this proposal are faux data composed of group members and consenting immediate family.

Business Rules

To identify the constraints and limitations of the system that will be simulated, the group has decided to formulate business rules and collect company rules from the Philippine Statistics Authority website. These rules will serve as the basis for how the system would be designed and the limits that will be set in order to maintain efficiency and integrity within the proposed system. The select rules from the Philippine Statistics Authority are classified under Company Rules while the rules formulated by the group are classified under Business Rules.

<table>
<tr><th>Company Rules</th><th>Company Rule Title</th></tr>
<tr><td>PSA-01</td><td>
<table>
<tr><th>registry_number</th><th>form_type</th><th>name</th></tr>
<tr><td>2026-1234567</td><td>Birth Certificate</td><td>Daitol, Michael Angelo</td></tr>
<tr><td>2026-1234568</td><td>Birth Certificate</td><td>Daitol, Michael Angelo</td></tr>
</table>
Registration Number Rule:
One registry number per certificate. A particular certificate may not have two separate registry numbers.
</td></tr>
</table>

PSA-02

registry_number	form_type
2026-0000001	**Birth Certificate**
2026-0000002	**Birth Certificate**
2026-0000010 (breaks continuity)	**Birth Certificate**

Indexing Rule:
Registry number must be continuous

PSA-03

Child Legal Personality Rule:
If a fetus has an intra-uterine life of less than seven (7) months it is not deemed born if it dies within twenty-four (24) hours after its complete delivery from the maternal womb (Article 41, R.A. 386). For statistical purposes, a Certificate of Live Birth (COLB) shall be prepared in duplicate, a copy shall be forwarded to the office of the Civil Registrar-General and the other for Civil Registrars file. (22:3a)

PSA-04

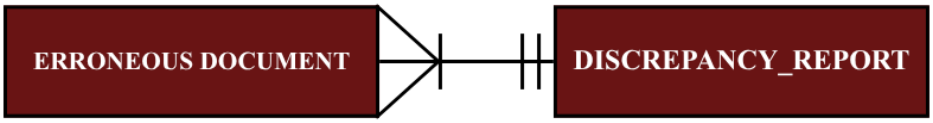
Date of Registration	Date of Certification
October 16, 2024	**October 14, 2024**
October 16, 2024	**November 28, 2024**

Date Validation Rule:
The Date of Registration must be later than or equal to the Date of Certification (Embalmer, Physician, or Clergy)

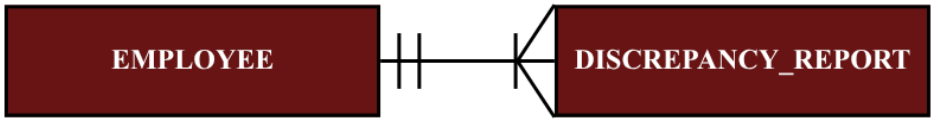
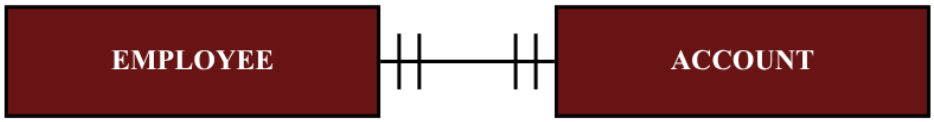
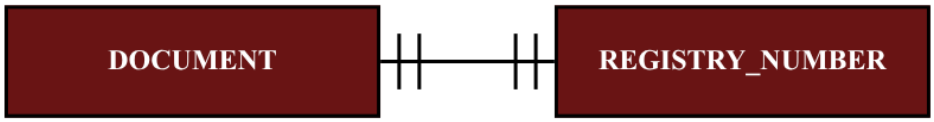
PSA-05

date_of_birth	Age
October 29, 2005	**Age 20**
October 29, 2005	**Age 21 (not full year has elapsed)**

Age Validation Rule:
The Age field shall represent the total number of full years elapsed between the Date of Birth and the Date of Encoding.

<table>
<tr><td>PSA-06</td><td>
<table>
<tr><th>registry_number</th><th>form_type</th></tr>
<tr><td>2026-1234567</td><td>Birth Certificate</td></tr>
<tr><td>2026-1234567</td><td>Marriage Certificate</td></tr>
<tr><td>2026-1234567</td><td>Death Certificate</td></tr>
</table>
Data Consistency Rule:
Multiple certificate types are permitted to share the same Registry Number, provided they pertain to the different records.
</td></tr>
<tr><td>PSA-07</td><td>Birth Name Exception Rule:
Birth names are not counted as errors in the context of what is deemed incorrect spellings and typographical errors unless the errors appear on certificates that are not a birth certificates</td></tr>
<tr><th colspan="2">Business Rules</th></tr>
<tr><th colspan="2">Entry Reporting System</th></tr>
<tr><td>BR-01</td><td>
allowed solutions per report


Discrepancy Report Limitation Rule (m:1)
While a discrepancy report may address discrepancies across various form types (e.g., Birth, Death, Marriage); however, each individual instance is restricted to resolving a single discrepancy per discrepancy report. Meaning that a document has many issues, each issue must be resolved as its own report.
</td></tr>
<tr><th colspan="2">Employee Rules</th></tr>
</table>

BR-03	**employees handling a report**  **Mandatory Employee Rule (1:m)** There must be at least one employee assigned to discrepancy reports. However; a report must be handled by exactly one employee
BR-04	**employees entitled to account**  **User Integrity Rule (1:1)** Each account given by the administration must not be used by multiple people. All accounts made are for exclusive use of the designated employee/user.
Birth Certificate Rules	
BR-05	**Registration Number assigned to a Document**  **Registration Number Integrity Rule (1:1)** Certificate registration numbers are constant on documents. A document cannot have changed or have multiple registration numbers.
Marriage Certificate Rules	

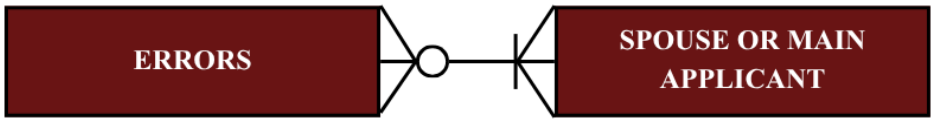
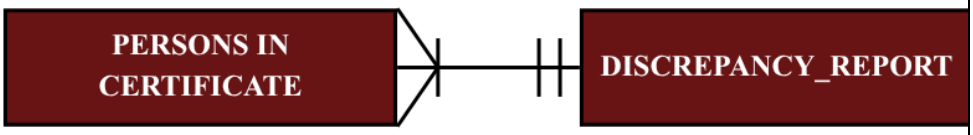
BR-06	**errors in a marriage certificate**  **Marriage Certificate Rule (m:n)** In regards to marriage certificates, errors must appear at least one party's field and may appear on both. However; there may be at least one error on each person in the marriage certificate
BR-07	**attribution of the report**  **Applicant Attribution Rule (1:1)** When a Marriage Certificate entry is being processed the person attributed between the spouses in the system is the one who has filed the complaint or the applicant only.

Table 1: Company and Business Rules

Problem Identification

Data discrepancies between physical copies of certificates and their corresponding digital records are a regular occurring issue in the workflow of the Philippine Statistics Authority (PSA). These discrepancies cause issues not only for the PSA, but for the owners of the actual document, as the process for correcting these mistakes is costly and tedious. Common discrepancies include: birth order mismatch in birth certificates; outdated age in marriage certificates; as well as the date of certification being earlier than the date of registration in death certificates. Such discrepancies may affect the integrity of these records, cause the delay on processing, and may lead to decreased productivity affecting both document owners and PSA personnel.

The current system that is used to resolve these discrepancies involves annotating the changes into a Microsoft Excel spreadsheet, manually recording each certificate's registry number and details of the corrected errors with the corrected data already inputted in the system. The spreadsheet serves as documentation for whenever employees are questioned as to why the physical copy has a different field value than the system. The central office then bears the responsibility to coordinate with the satellites regarding the revisions the employees have conducted. Without a centralized system that intends to manage these discrepancies, revisions are verified through other forms of communications such as phone calls to investigate the reason for such revisions. This current process is tedious and time consuming for both satellite and central office personnel. Therefore a centralized database system is needed to improve the process of managing discrepancies and corrections. Implementing such a database system can improve productivity by centralizing reports and verification processes, while maintaining data integrity and improving the general workflow.

Objective of the Study/System

As such issues linger within the systems of the Philippine Statistics Authority, the researchers have devised the following objectives that serve as guidelines to achieve, in order to simulate and improve upon the error-collection systems. With the main goal of modernizing error-handling architecture, the proposed system is designed to meet the following functional requirements:

- A simulation of an error-collection system that processes and stores data amongst faulty certificate records.
- Presents a persistent audit log that provides traceability of all user activities and record modifications.
- Providing a secure authentication layer employing a verified login procedure and a restrictive system for authorized users only.

FORMS AND TABLES

The group has not been given any forms to work with as this process of storing and correcting erroneous data was done manually through a Microsoft Excel sheet that is off-limits to the group due to confidentiality reasons. However, presented in the Normalization section is the Unnormalized Form for the table created by the group which closely approximates the look and function of the Excel sheet used by the PSA which can be viewed in figure 1.1.

NORMALIZATION

The process of normalization is done to condense the database system's entities and attributes such that data management may be optimized by means of removing many forms of redundancy in the data and organizing the tables such that the relations may still uphold while reducing the opportunity for data anomalies to exist. The following section presents the full process of normalization on the database entities and attributes of the group's table, starting from Unnormalized Form (UNF) until the most normalized 3NF table, which will be the basis for the group's data dictionary and Entity Relationship Table (ERD).

UNF (Unnormalized Form)

This section is to illustrate the initial state of the database table; the unnormalized form is the most primitive and unsorted form of the table system. Exhibiting redundancies and characterized by normalization violations, as well as the existence of composite–or grouped attributes, widespread NULL values, and partial and trivial dependencies issues remain. While the full table is displayed to show the entirety, the specific portions of the table are emphasized for visibility and analysis of structural change, with reference to the entire table.

UNF



Figure 1.1: Full Unnormalized Form of the Database System

report_id	date_issued	registry_no	form _type
BC-26-04-12345	27/04/2026	2026-1234567	Birth certificate
BC-26-04-12346	28/04/2026	2026-1234345	Death Certificate
BC-26-04-12347	28/04/2026	2026-1233667	Marriage Certificate
BC-26-04-12348	29/04/2026	2026-2468101	Birth certificate
BC-26-04-12349	30/04/2026	2026-3691215	Birth certificate
BC-26-05-12350	01/05/2026	2026-2905177	Marriage Certificate
BC-26-05-12351	7/5/2026	2026-3023432	Marriage Certificate
BC-26-05-12352	10/5/2026	2026-1017005	Death Certificate

Figure 1.2: Report Category of the UNF Table

nameof_child	dob	father_name	mother_name	birth_order
Liu, Landon Riley	25/11/2020	Liu, Roland Pacio	Concepcion, Ina Racel Ferrer	1
Francisco, Marl Andrei Tagle	29/10/2012	Francisco, Marlon Hadap	Tagle, Angelyn Flores	2
Pangilinan, Ezekiel Balais	12/11/2005	Pangilinan, Erwin Barro	Balais, Mary Grace Toledo	1

Figure 1.3: Birth Certificate Category of the UNF Table

deceased_name	birth_date	death_date	date_registration	date_certification
Pangilinan, Ezekiel B.	12/11/2005	14/11/2005	16/11/2005	15/11/2005
Concepcion, Ramcel Aaron Ferrer	17/10/2025	17/10/2021	21/10/2021	19/10/2021

Figure 1.4: Death Certificate Category of the UNF Table

applicant_name	applicant_age	spouse_name	spouse_age
Daitol, Michael Angelo	21	Francisco, Marl Louie	20
Concepcion, Ramiro Fonseca	59	Concepcion, Carina Celeste Ferrer	58
Liu, Ina Racel Concepcion	33	Liu, Roland Pacio	34

Figure 1.4: Marriage Certificate Category of the UNF Table

workaround_id	error_field_name	Error_value	Solution_value	employee_id	employee_name	ongoing status
E-0200	birth_order	3	1	EMP-005	Concepcion, Ramcel Aaron	resolved
E-0201	date_registration	11/13/2026	11/16/2026	EMP-006	Biando, Sofia Ela	resolved
E-0202	age	23	21	EMP-006	Biando, Sofia Ela	unresolved
E-0203	birth_order	3	2	EMP-005	Concepcion, Ramcel Aaron	resolved
E-0204	dob	11/11/2005	12/11/2005	EMP-007	Francisco, Marl Louie	unresolved
E-0205	applicant_age	58	59	EMP-007	Francisco, Marl Louie	unresolved
E-0206	spouse_name	Liu, Rolan Pacio	Liu, Roland Pacio	EMP-008	Sisante, Humphrey	unresolved
E-0207	date_regitration	22/10/2021	21/10/2021	EMP-007	Francisco, Marl Louie	resolved

Figure 1.5: Error-Correction Category of the UNF Table

1NF (1st Normal Form)

First Normal Form in context is the process of establishing an initial but unfinalized structured schema entities and attributes, eliminating redundancies by removing the amount of composite values into atomic, individual values. Specifically, the general 'name' attribute is subdivided into first, middle, and last name; along with the birthdate decomposed into birth year, month, and day. Sustaining a structure amongst table fields, and the addition of unique identifiers for columns and rows; moreover, removing repeating groups from the list. This transformation to 1NF effectively creates a more logical and coherent form than the unnormalized form.

Birth Certificate

registry_number (PK)	child_last_name	child_middle_name	child_first_name	birth_year	birth_month
2026-1234567	Liu	Concepcion	Landon Riley	2020	11
2026-2468101	Francisco	Tagle	Marl Andrei	2012	10
2026-3691215	Pangilinan	Balais	Ezekiel	2005	11

birth_day	mother_maiden_last_name	mother_maiden_middle_name	mother_first_name	father_last_name	father_middle_name
25	Concepcion	Ferrer	Ina Racel	Liu	Pacio
29	Tagle	Flores	Angelyn	Francisco	Hadap
12	Balais	Toledo	Mary Grace	Pangilinan	Barro

father_first_name	birth_order
Roland	1
Marlon	2
Erwin	1

Figure 2.1: Birth Certificate Entity Normalized to 1st Normal Form

Marriage Certificate

registry_number (PK)	applicant_last_name	applicant_middle_name	applicant_first_name	applicant_birth_year	applicant_birth_month
2026-2905177	Concepcion	Fonseca	Ramiro	1966	5
2026-3023432	Liu	Concepcion	Ina Racel	1993	1

applicant_birth_day	applicant_age	spouse_last_name	spouse_middle_name	spouse_first_name	spouse_birth_year
5	59	Concepcion	Ferrer	Carina Celeste	1967
29	33	Liu	Pacio	Roland	1992

spouse_birth_month	spouse_birth_day	spouse_age
11	7	58
5	4	34

Figure 2.2: Marriage Certificate Entity Normalized to 1st Normal Form

Death Certificate

registry_number (PK)	deceased_last_name	deceased_middle_name	deceased_first_name	birth_year	birth_month
2026-1234345	Pangilinan	Balais	Ezekiel	2005	11
2026-1017005	Concepcion	Ferrer	Ramcel Aaron	2005	10

birth_day	death_year	death_month	death_day	deceased_age	registration_year	registration_month
12	2005	11	14	0	2005	11
17	2021	10	17	16	2021	10

registration_day	certification_year	certification_month	certification_day
15	2005	11	16
19	2021	10	21

Figure 2.3: Death Certificate Entity Normalized to 1st Normal Form

Workaround

workaround_id	report_id	explanation	field_name	original_value	revised_value
E-0200	BC-26-04-12345	incorrect order of birth	birth_order	3	1
E-0201	BC-26-04-12346	incosistent date of registration	date_registration	11/13/2026	11/16/2026
E-0202	BC-26-04-12347	outdated age	age	23	21
E-0203	BC-26-04-12348	incorrect order of birth	birth_order	3	2
E-0204	BC-26-04-12349	incorrect date of birth	dob	11/11/2005	12/11/2005
E-0205	BC-26-04-12350	outdated age	applicant_age	58	59
E-0206	BC-26-04-12351	incorrect name	spouse_name	Liu, Rolan Pacio	Liu, Roland Pacio
E-0207	BC-26-04-12352	inconsistent date of registrat	date_regitration	22/10/2021	21/10/2021

Figure 2.4: Error-Correction Entity Normalized to 1st Normal Form

Discrepancy Report

report_id (PK)	emp_id	registry_number	form_type	status
BC-26-04-12345	EMP-005	2026-1234567	Birth certificate	resolved
BC-26-04-12346	EMP-006	2026-1234345	Death Certificate	resolved
BC-26-04-12347	EMP-006	2026-1233667	Marriage Certificate	unresolved
BC-26-04-12348	EMP-005	2026-2468101	Birth certificate	resolved
BC-26-04-12349	EMP-007	2026-3691215	Birth certificate	unresolved
BC-26-05-12350	EMP-007	2026-2905177	Marriage Certificate	unresolved
BC-26-05-12351	EMP-008	2026-3023432	Marriage Certificate	unresolved
BC-26-05-12352	EMP-007	2026-1017005	Death Certificate	resolved

Figure 2.5: Discrepancy Report Entity Normalized to 1st Normal Form

Employee

employee_id	emp_last_name	emp_middle_name	emp_first_name
EMP-005	Concepcion	Ferrer	Ramcel Aaron
EMP-006	Biando	Evangelista	Sofia Ela
EMP-007	Francisco	Tagle	Marl Louie
EMP-008	Sisante	Revilla	Humphrey

Figure 2.6: Employee Entity Normalized to 1st Normal Form

2NF (2nd Normal Form)

The 2nd Normal Form (2NF) version of the table takes the previous normal form of the entities and attributes and removes any partial dependencies that may contain redundant connections. This step removes the non-key attributes are dependent fully on the primary and not partially.

BIRTH_CERTIFICATE

registry_number(PK)	child_last_name	child_middle_name	child_first_name
2026-1234567	Liu	Concepcion	Landon Riley
2026-2468101	Francisco	Tagle	Marl Andrei
2026-3691215	Pangilinan	Balais	Ezekiel

birth_year	birth_month	birth_day
2020	11	25
2012	10	29
2005	11	12

mother_maiden_last_name	mother_maiden_middle_name	mother_first_name
Concepcion	Ferrer	Ina Racel
Tagle	Flores	Angelyn
Balais	Toledo	Mary Grace

father_last_name	father_middle_name	father_first_name	birth_order
Liu	Pacio	Roland	1
Francisco	Hadap	Marlon	2
Pangilinan	Barro	Erwin	1

Figure: 3.1: Birth Certificate Entity normalized to 2nd Normal Form

MARRIAGE_CERTIFICATE

registry_number (PK)	applicant_last_name	applicant_middle_name	applicant_first_name
2026-2905177	Concepcion	Fonseca	Ramiro
2026-3023432	Liu	Concepcion	Ina Racel

applicant_birth_year	applicant_birth_month	spouse_first_name
1966	5	5
1993	1	29

spouse_last_name	spouse_middle_name	spouse_first_name
Concepcion	Ferrer	Carina Celeste
Liu	Pacio	Roland

spouse_birth_year	spouse_birth_month
1967	11
1992	5

Figure: 3.2: Marriage Certificate Entity normalized to 2nd Normal Form

DEATH_CERTIFICATE

registry_number (PK)	deceased_last_name	deceased_middle_name,	deceased_first_name,	birth_year,
2026-1234345	Pangilinan	Balais	Ezekiel	2005
2026-1017005	Concepcion	Ferrer	Ramcel Aaron	2005

birth_month	birth_day	death_year	death_month	death_day	deceased_age
11	12	2005	November	14	0
10	17	2021	October	17	16

registration_month	registration_day	registration_year	certification_month	certification_day	certification_year
11	16	2005	11	18	2005
10	19	2021	10	21	2021

Figure: 3.3: Death Certificate Entity normalized to 2nd Normal Form

DISCREPANCY_REPORT

report_id (PK)	employee_id (FK)	registry_number (FK)	record_type (Birth/Marriage/Death)	status
BC-26-04-12345	EMP-005	2026-1234567	Birth certificate	resolved
BC-26-04-12346	EMP-006	2026-1234345	Death Certificate	resolved
BC-26-04-12347	EMP-006	2026-1233667	Marriage Certificate	unresolved
BC-26-04-12348	EMP-005	2026-2468101	Birth certificate	resolved
BC-26-04-12349	EMP-007	2026-3691215	Birth certificate	unresolved
BC-26-05-12350	EMP-007	2026-2905177	Marriage Certificate	unresolved
BC-26-05-12351	EMP-008	2026-3023432	Marriage Certificate	unresolved
BC-26-05-12352	EMP-007	2026-1017005	Death Certificate	resolved

Figure: 3.4: Discrepancy Report Entity normalized to 2nd Normal Form

WORKAROUND_ENTRY

workaround_id (PK)	report_id (FK)	explanation	error_field
E-0200	BC-26-04-12345	incorrect order of birth	birth_order
E-0201	BC-26-04-12346	incosistent date of registration	date_registration
E-0202	BC-26-04-12347	outdated age	age
E-0203	BC-26-04-12348	incorrect order of birth	birth_order
E-0204	BC-26-04-12349	incorrect date of birth	date of birth
E-0205	BC-26-04-12350	outdated age	applicant_age
E-0206	BC-26-04-12351	incorrect name	spouse_name
E-0207	BC-26-04-12352	inconsistent date of registration	date_regitration

original_value	revised_value	employee_id (PK)	emp_last_name	emp_middle_name	emp_first_name
3	1	EMP-005	Concepcion	Ferrer	Ramcel Aaron
11/13/2026	11/16/2026	EMP-006	Biando	Evangelista	Sofia Ela
23	21	EMP-006	Biando	Evangelista	Sofia Ela
3	2	EMP-005	Concepcion	Ferrer	Ramcel Aaron
11/11/2005	12/11/2005	EMP-007	Francisco	Tagle	Marl Louie
58	59	EMP-007	Francisco	Tagle	Marl Louie
Liu, Rolan Pacio	Liu, Roland Pacio	EMP-008	Sisante	Revilla	Humphrey
22/10/2021	21/10/2021	EMP-007	Francisco	Tagle	Marl Louie

Figure: 3.5: Error-Correction Entry Entity normalized to 2nd Normal Form

3NF (3rd Normal Form)

The 3rd Normal Form (3NF) stage is the last in the normalization process as this stage ensures that each attribute within entities is reliant on only one primary key, not transitive connections. This is done by taking the 2nd Normal Form (2NF) and removing all transitive dependencies through ensuring that all non-prime attributes are not reliant on other non-prime attributes.

PERSON

person_id (PK)	last_name	first_name	middle_name	birth_date
P-10010	Liu	Landon Riley	Concepcion	11/25/2020
P-10021	Francisco	Marl Andrei	Tagle	10/29/2012
P-10022	Pangilinan	Ezekiel	Balais	11/12/2005
P-10023	Concepcion	Ramiro	Fonseca	5/5/1966
P-10024	Liu	Ina Racel	Concepcion	1/29/1993
P-10025	Pangilinan	Ezekiel	Balais	11/12/2005
P-10026	Concepcion	Ramcel Aaron	Ferrer	10/17/2005
P-10027	Liu	Roland	Pacio	5/4/1993
P-10029	Concepcion	Carina Celeste	Ferrer	11/7/1967
P-10030	Daitol	Guardy	Boyles	2/28/1997
P-10031	Daitol	Michael Angelo	Alcantara	10/2/2005
P-10032	Alcantara	Jaqueline	Macalam	11/8/1989
P-10033	Biando	Sofia Ela	Evangelista	7/3/2006

Figure 4.1: Person Entity Normalized to 3rd Normal Form

EMPLOYEE

employee_id	last_name	first_name	middle_name
ramcelaaron	Concepcion	Ramcel Aaron	Ferrer
ruiiumaru	Francisco	Marl Louie	Tagle
sofiaela	Biando	Sofia Ela	Evangelista
ezekiel	Pangilinan	Ezekiel	Balais
alto	Sisante	Humphrey	Revilla
admin	admin	admin	admin

Figure 4.2: Employee Entity Normalized to 3rd Normal Form

EMPLOYEE_ACCOUNT

employee_id	email	password	created_datetime	modified_datetime
ramcelaaron	nramcelaaron@	ramcelaaron	2026-01-10	2026-01-17
ruiiumaru	e.francisco62@g	ruiiumaru	2026-01-24	2026-01-29
sofiaela	iaela.06@gmail.c	sofiaela	2026-03-15	2026-03-31
ezekiel	ngilinan13@yaho	ezekiel	2026-03-20	2026-03-21
alto	Alto@psa.gov.ph	111	2026-03-20	2026-03-21
admin	dmin@gmail.com	admin123	2026-01-01	2026-01-01

Figure 4.2.1: Employee Account Entity Normalized to 3rd Normal Form

BIRTH CERTIFICATE

registry_number (PK)	child_id (FK -> PERSON)
2026-1234567	P-10010
2026-2468101	P-10026

Figure 4.3: Birth Certificate Entity Normalized to 3rd Normal Form

MARRIAGE_CERTIFICATE

registry_number (PK)	applicant_id (FK -> PERSON)
2026-2905177	P-10023
2026-3023432	P-10024

Figure 4.4: Marriage Certificate Entity Normalized to 3rd Normal Form

DEATH_CERTIFICATE

registry_number (PK)	deceased_id (FK -> PERSON)
2026-1234345	P-10025
2026-1017005	P-10026

Figure 4.5: Death Certificate Entity Normalized to 3rd Normal Form

DISCREPANCY_REPORT

report_id (PK)	person_name	cert_type	registry_number	status	explanation
26-04-12345	Concepcion, Landon R	Birth Certificate	2026-1234567	resolved	incorrect order of birth
26-04-12346	Concepcion, Ramiro Fc	Marriage Certficate	2026-2905177	resolved	incosistent date of registration
26-04-12347	Pangilinan, Ezekiel Bal	Death Certificate	2026-1233667	unresolved	outdated age
26-04-12348	Francisco, Marl Andrei	Birth Certificate	2026-2468101	resolved	incorrect order of birth
26-04-12349	Pangilinan, Ezekiel Bal	Birth Certificate	2026-3691215	unresolved	incorrect date of birth
26-05-12350	Concepcion, Ramiro Fc	Marriage Certficate	2026-2905177	unresolved	outdated age
26-05-12351	Liu, Ina Racel Concepc	Marriage Certficate	2026-3023432	unresolved	incorrect name
26-05-12352	Concepcion, Ramcel A	Death Certificate	2026-1017005	resolved	inconsistent date of registration

error_field	original_value	revised_value	created_by	modified_by	created_date	modified_date
birth_order	3	1	ramcelaaron	ramcelaaron	4/12/2026 14:34:12	4/13/2026 9:18:27
date_registration	11/13/2026	11/16/2026	ruiiumaru	ruiiumaru	4/15/2026 10:22:44	4/16/2026 16:07:55
age	23	21	sofiaela	sofiaela	4/18/2026 8:11:30	4/18/2026 8:11:30
birth_order	3	2	ezekiel	ezekiel	4/20/2026 13:45:09	4/22/2026 11:32:14
date_of_birth	11/11/2005	12/11/2005	alto	alto	4/27/2026 15:56:41	4/27/2026 15:56:41
applicant_age	58	59	ezekiel	ezekiel	5/4/2026 9:27:36	5/4/2026 9:27:36
spouse_name	Liu, Rolan Pacio	Liu, Roland Pacio	alto	alto	5/12/2026 16:42:18	5/12/2026 16:42:18
date_regitration	22/10/2021	21/10/2021	sofiaela	admin	5/19/2026 11:08:53	5/20/2026 14:55:31

Figure 4.6: Discrepancy Report Entity Normalized to 3rd Normal Form

ENTITY RELATIONSHIP DIAGRAM

Presented here is the entity relationship diagram of the proposed database system. The entities and attributes were refined through the normalization process as shown in the previous section of this proposal and given properly identified relations inline with the process of the intended system.

Before presenting the relations of the entities to each other, here is a brief description of each entity.

- ***PERSON*** is the center of the other entity types as it contains the record for every person in the collection. It contains the names, birth and ID that are borrowed by the other entities.
- ***EMPLOYEE*** is a specific denomination of PERSON as these are the records of people who oversee or file the reports for data correction. These are necessary entities for the system to function.

- ***EMPLOYEE_ACCOUNT*** is the account that the EMPLOYEE is to use when entering error reports
- ***EMPLOYEE_ACCOUNT_LOGS*** monitors the activities that occur surrounding the account to check for any anomalies in the account creation and modification
- ***FORMS*** is not an entity; however, for the sake of definition, it is the classification of the form type that is passed to the DISCREPANCY_REPORT entity whenever a report is filed. It concerns the entities ***DEATH_CERTIFICATE, MARRIAGE_CERTIFICATE, or BIRTH_CERTIFICATE***.
- ***DISCREPANCY_REPORT*** is the entity/process of reporting on any erroneous data or anomalies.
- ***DISCREPANCY_REPORT_AUDIT*** is the collecting entity for DISCREPANCY_REPORT

The relation between the aforementioned entities also illustrates the process of the entry system which is shown in the ERD graph. The Entity Relationship Diagram discloses the following:

I. Every single certificate and employee entity is linked to the PERSON entity as this is where the basic data of the person (name, ID) are stored.
II. There must always be at least one EMPLOYEE because a report cannot exist without an employee handling the filed report to the DISCREPANCY_REPORT.
III. Every EMPLOYEE has an EMPLOYEE ACCOUNT to access the main entry system and thus also have an EMPLOYEE_ACCOUNT_LOG to keep track of things such as password updates, username changes and other security monitoring matters.
IV. BIRTH_CERTIFICATE reports require three person_id foreign keys to supplement the needs of the child, mother, and father IDs.
V. MARRIAGE_CERTIFICATE reports require two person_id foreign keys to supplement the needs of the applicant_id and the spouse_id.

VI. each FORM may be sent many reports; however, each form may only be attributed to only one report perf form.

VII. The DISCREPANCY_REPORT must send at least one report; however, each report must only get one solution

VIII. The DISCREPANCY_REPORT_AUDIT is the entity that handles the storage of all the reports and their properties for audit logs.

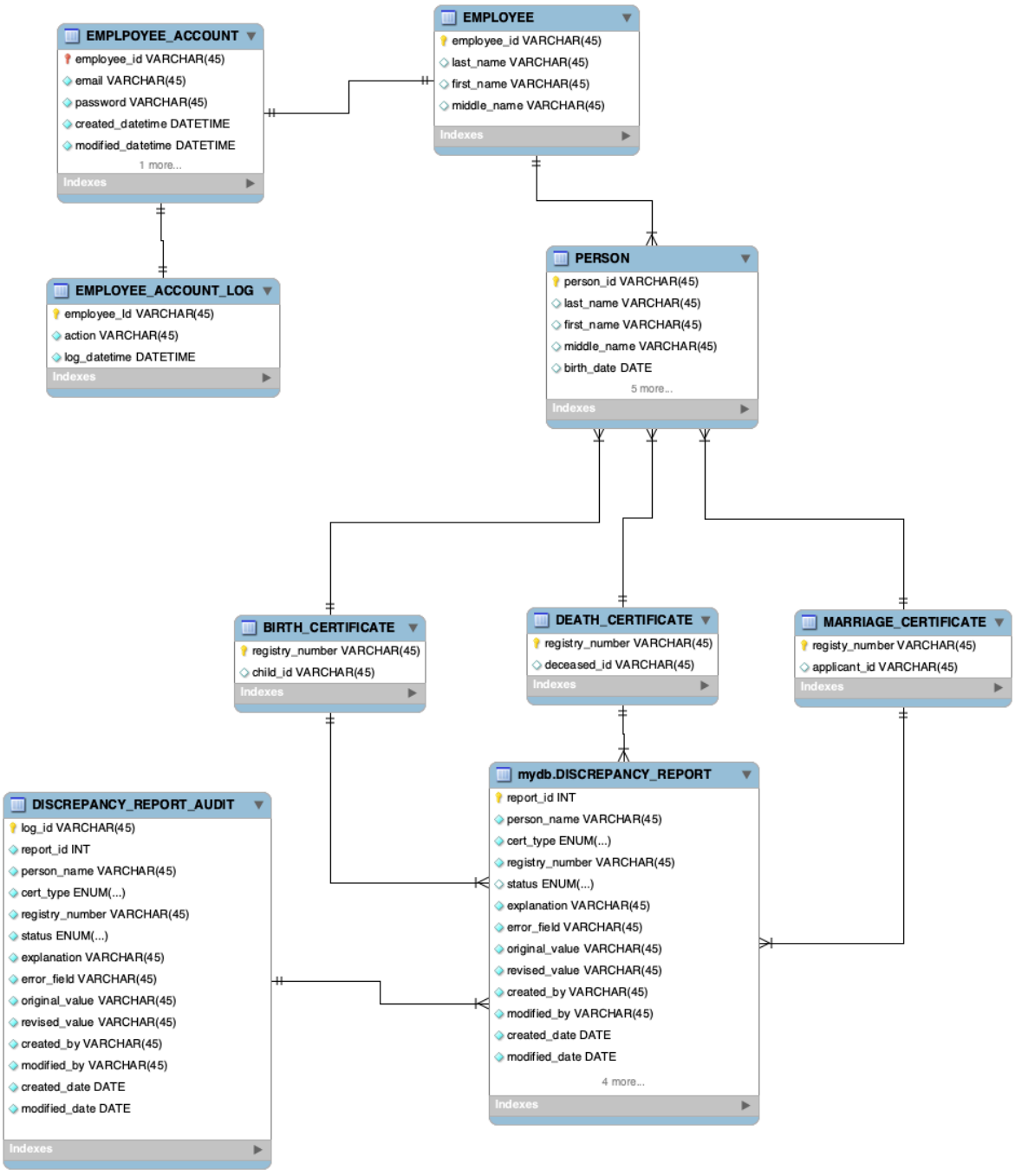


Figure 5.1: Entity Relationship Diagram of the Database System

ENHANCED ENTITY RELATIONSHIP DIAGRAM

The Enhanced Entity Relationship Diagram is a more expanded version of the ERD shown prior with the integration of object-oriented concepts such as inheritance, class systems, and other mechanisms to further explain the mechanics of the system.

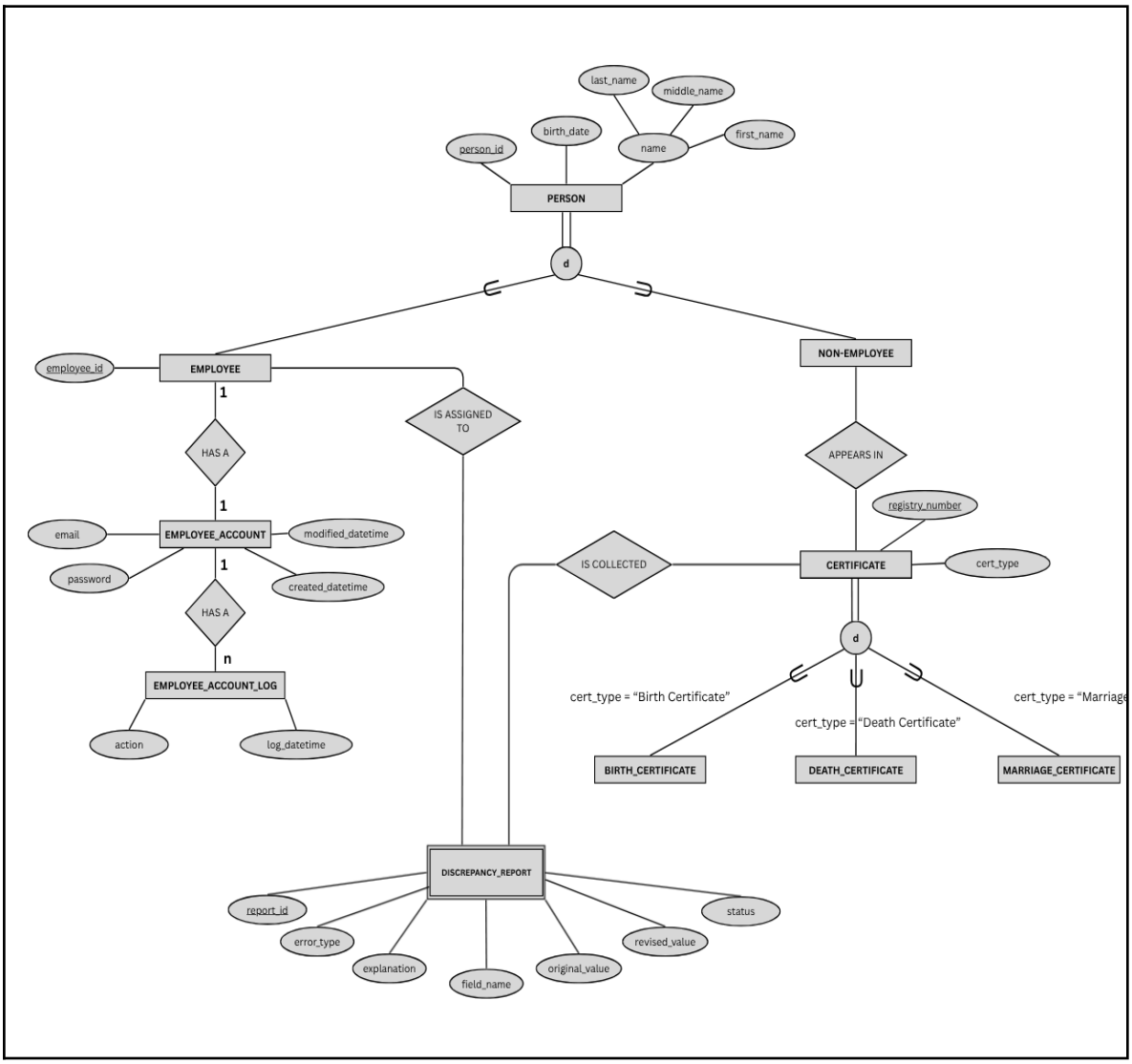


Figure 5.2: Enhanced Entity Relationship Diagram of the Database System

The diagram shows how the person attribute is the divider between employees and non-employees who are to appear in certificates as names. which moves down their respective fields with employees passing down their information towards the account all

the way to the logs. the non-employee is set as a name in the certificates granted they meet the specifier to distinguish the types of certificate used through the “cert_type” attribute. Lately, the dynamic between the certificate and employee is illustrated by the diamond relationship connectors being is assigned to an entry and is collected as an entry.

DATA DICTIONARY

This data dictionary is the section of the proposal where the proper names and their respective definitions are added. In this section, the group has included all of the entities and attributes of the database system which is taken from the finalized 3rd Normal Form version of the table. Included also are the details surrounding data type and the particular constraint given to each which will apply to the system and its functions.

PERSON

Field Name	Description	Type	Constraints
person_id	Unique identifier for a person	VARCHAR	PK
first_name	Given name	VARCHAR	NOT NULL
middle_name	Middle name	VARCHAR	NULL
last_name	Family name	VARCHAR	NOT NULL
birth_date	Date of birth	DATE	NOT NULL

BIRTH CERTIFICATE

Field Name	Description	Type	Constraints
registry_number	Unique birth record ID	VARCHAR	PK
child_id	Child person reference	VARCHAR	FK -> PERSON

MARRIAGE_CERTIFICATE

Field Name	Description	Type	Constraints
registry_number	Unique marriage record ID	VARCHAR	PK
applicant_id	Person who filed the	VARCHAR	FK -> PERSON

	record		

DEATH_CERTIFICATE

Field Name	Description	Type	Constraints
registry_number	Unique death record ID	VARCHAR	PK
deceased_id	Deceased person	VARCHAR	FK -> PERSON

DISCREPANCY_REPORT

Field Name	Description	Type	Constraints
report_id	Unique report ID	VARCHAR	PK
person_name	Name of certificate owner	VARCHAR	NOT NULL
cert_type	Type of certificate (*'Birth Certificate', 'Death Certificate', 'Marriage Certificate'*)	ENUM	NOT NULL
registry_number	Unique record ID of the certificate	VARCHAR	FK, NOT NULL
status	Report status (*'PENDING', 'RESOLVED'*)	ENUM	NOT NULL
explanation	Explanation of the fix	VARCHAR	NOT NULL
error_field	Affected field	VARCHAR	NOT NULL
original_value	Old value	VARCHAR	NOT NULL
revised_value	New value	VARCHAR	NOT NULL
created_by	The employee who made the report	VARCHAR	FK -> EMPLOYEE
modified_by	Employee ID of who modified the report.	VARCHAR	NOT NULL
created_datetime	Date and time of when	DATETIME	NOT NULL

	the report was published.		
modified_datetime	Same date and time as created_date until modified. (eg. status update)	DATETIME	NOT NULL

DISCREPANCY_REPORT_AUDIT

Field Name	Description	Type	Constraints
log_id	Unique ID. The old version of the report is saved in this entity.	INT	PK, Auto Increment
report_id	Unique report ID	VARCHAR	PK
person_name	Name of certificate owner	VARCHAR	NOT NULL
cert_type	Type of certificate (*'Birth Certificate', 'Death Certificate', 'Marriage Certificate'*)	ENUM	NOT NULL
registry_number	Unique record ID of the certificate	VARCHAR	FK, NOT NULL
status	Report status (*'PENDING', 'RESOLVED'*)	ENUM	NOT NULL
explanation	Explanation of the fix	VARCHAR	NOT NULL
error_field	Affected field	VARCHAR	NOT NULL
original_value	Old value	VARCHAR	NOT NULL
revised_value	New value	VARCHAR	NOT NULL
created_by	The employee who made the report	VARCHAR	FK -> EMPLOYEE
modified_by	Employee ID of who modified the report.	VARCHAR	NOT NULL
created_date	Date of when the report	DATETIME	NOT NULL

	was published.		
modified_date	Same date as created_date until modified. (eg. status update)	DATETIME	NOT NULL
audit_timestamp	Date and time of when the audit log is saved.	DATETIME	NOT NULL

EMPLOYEE

Field Name	Description	Type	Constraints
employee_id	Unique ID	VARCHAR	PK
last_name	Family name	VARCHAR	NOT NULL
first_name	Given name	VARCHAR	NOT NULL
middle_name	Middle name	VARCHAR	NULL

EMPLOYEE_ACCOUNT

Field Name	Description	Type	Constraints
employee_id	Unique ID	VARCHAR	PK
email	Employee Email	VARCHAR	NOT NULL
password	Password	VARCHAR	NOT NULL
created_datetime	Date and time of account creation.	DATETIME	NOT NULL
modified_datetime	Date and time of account modification.	DATETIME	NOT NULL

EMPLOYEE_ACCOUNT_LOGS

Field Name	Description	Type	Constraints
employee_id	Unique ID	VARCHAR	PK
action	Action performed by account. (eg. “account created”, “change	VARCHAR	NOT NULL

	password")		
log_datetime	Date and time of account modification.	DATETIME	NOT NULL

PROTOTYPES

Prototype (Low fidelity)

The group has created an illustration of the desired user interface for the system. The UI has been designed to provide convenience and efficiency through its simple and understandable design.

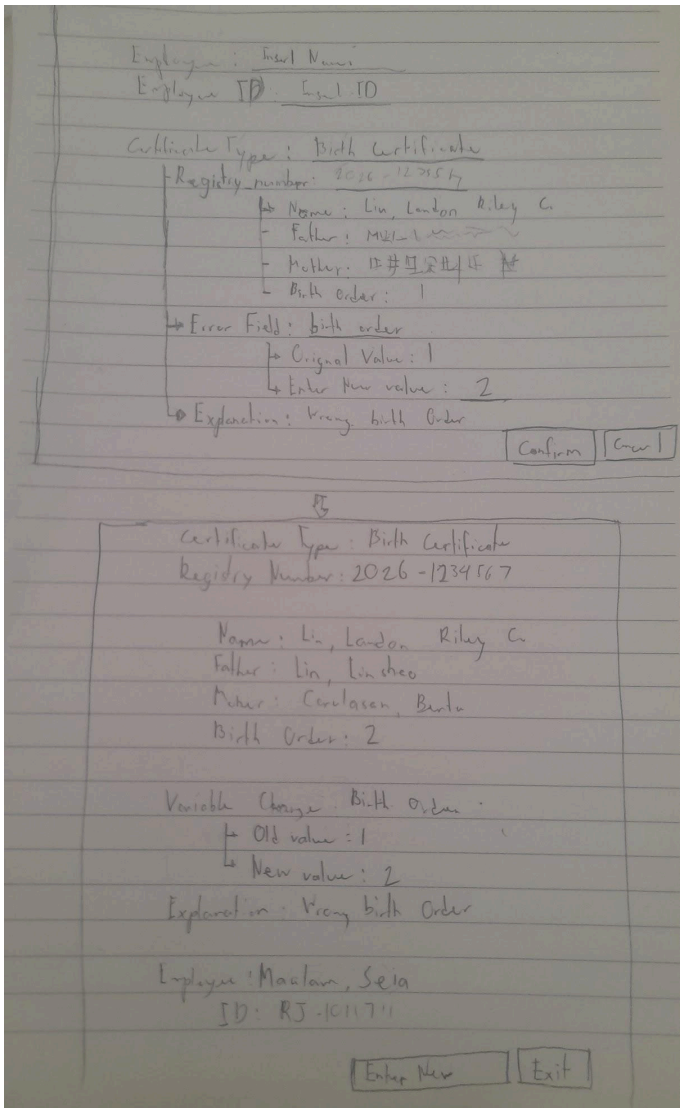


Figure 6: Low Fidelity Pencil Drawn UI

Screenshots of UI

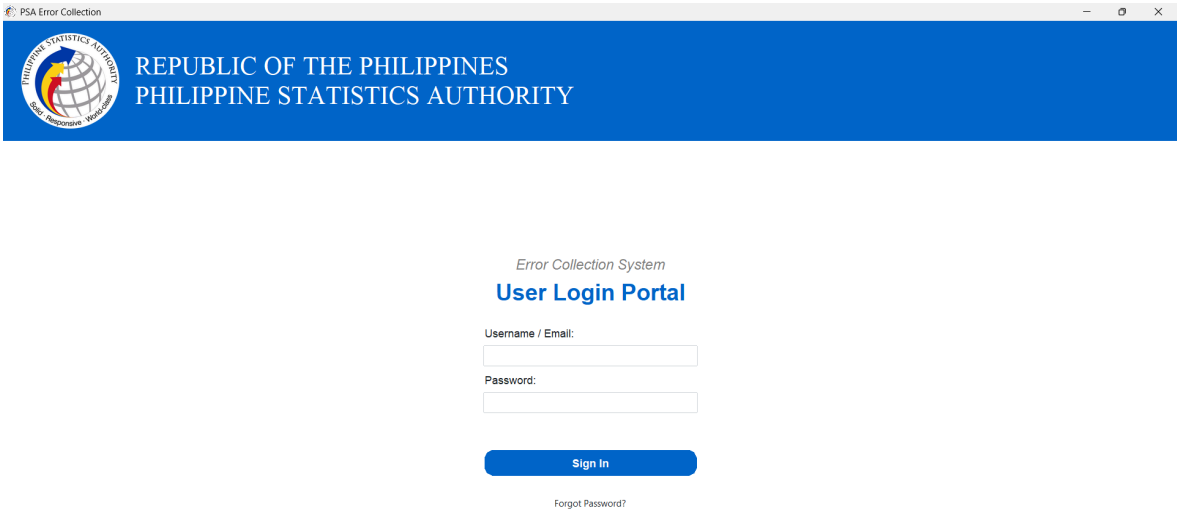


Figure 7: User Login Portal

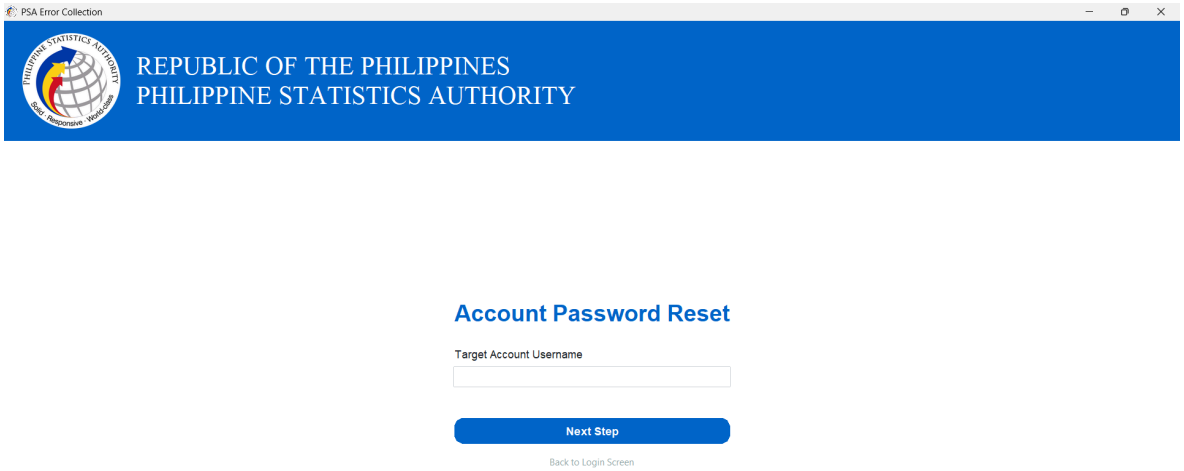


Figure 8: Reset Password Portal

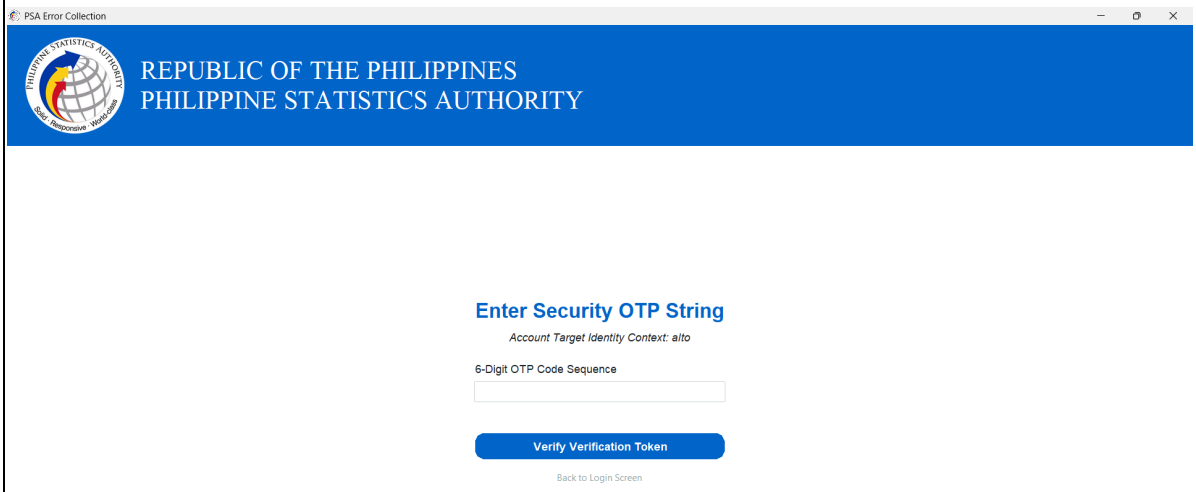


Figure 9: Reset Password Verification

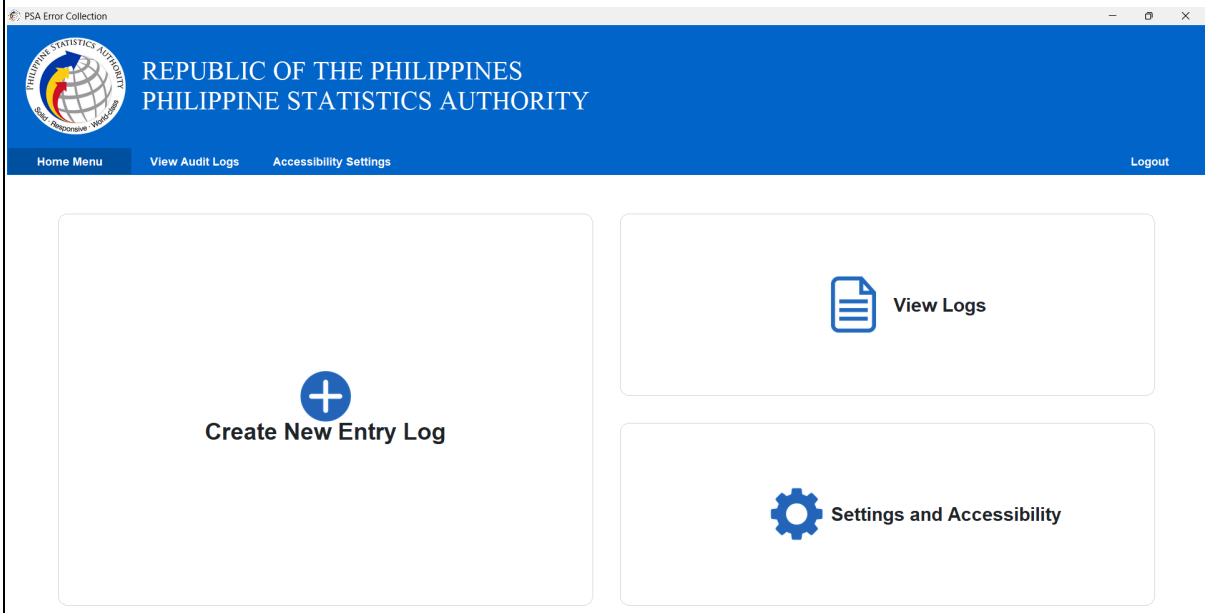


Figure 10: Employee-end Main Interface



Figure 11: Main Discrepancy Entry Interface

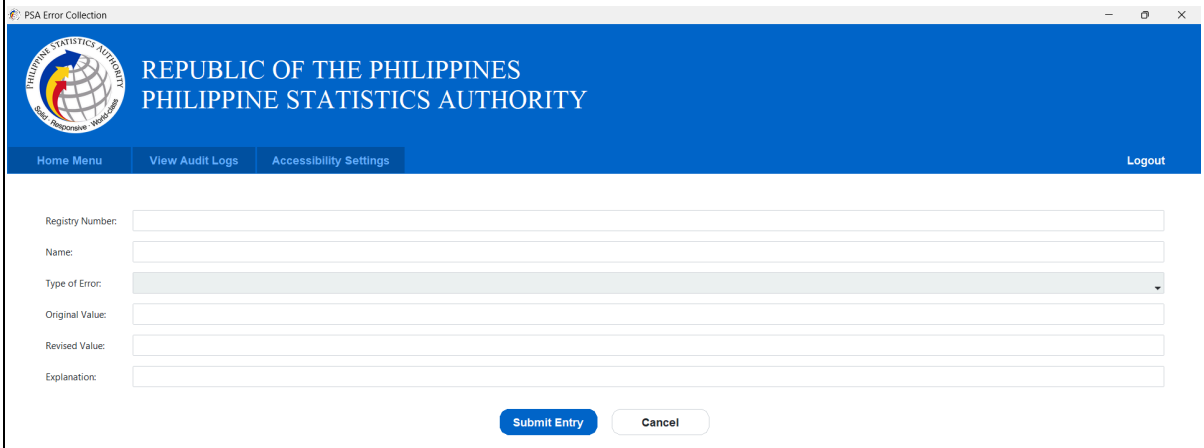


Figure 12: Entry Form for Birth Certificate

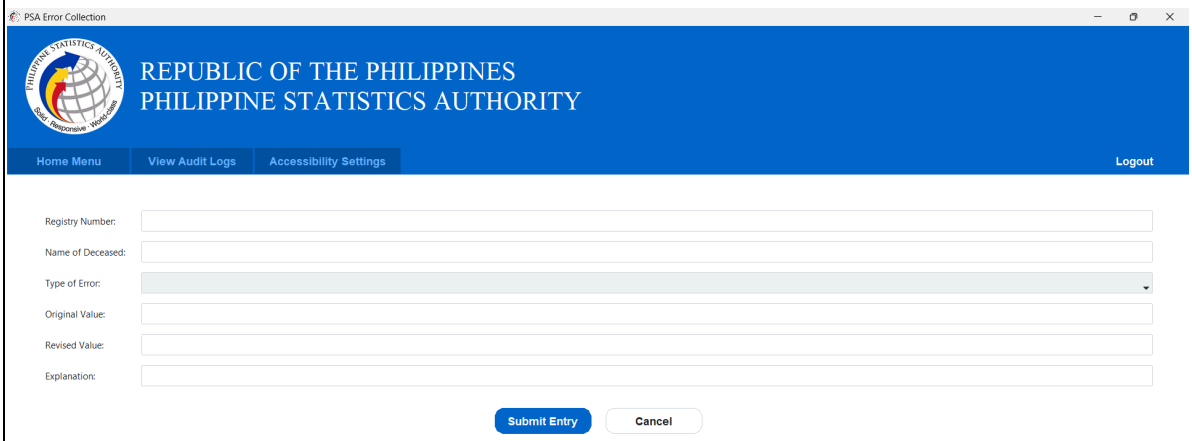


Figure 13: Entry Form for Death Certificate

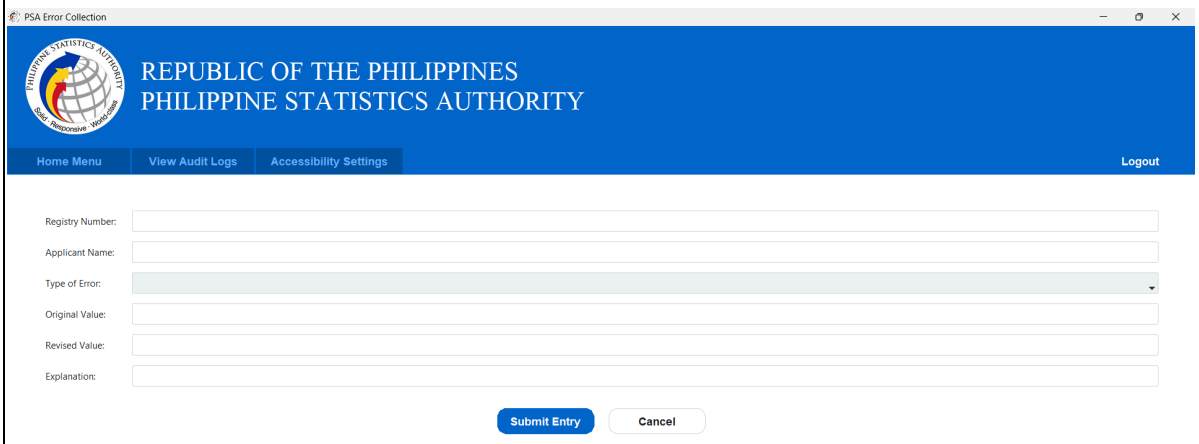


Figure 14: Entry Form for Marriage Certificate



Figure 15: Audit Log Main Interface

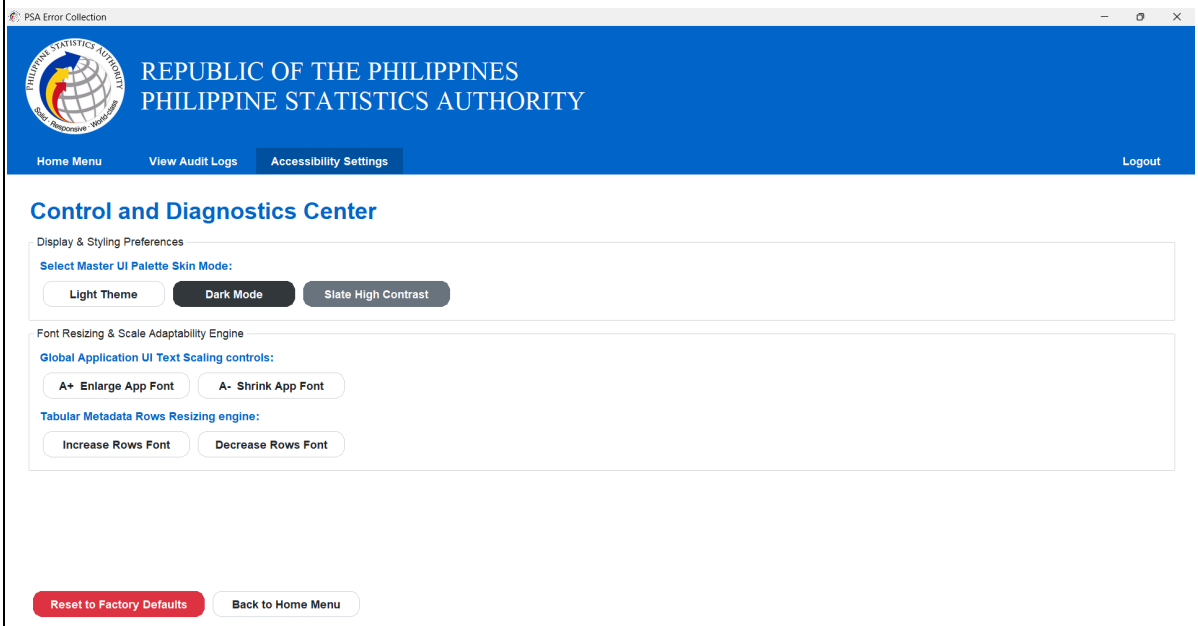


Figure 16: Accessibility Settings Interface

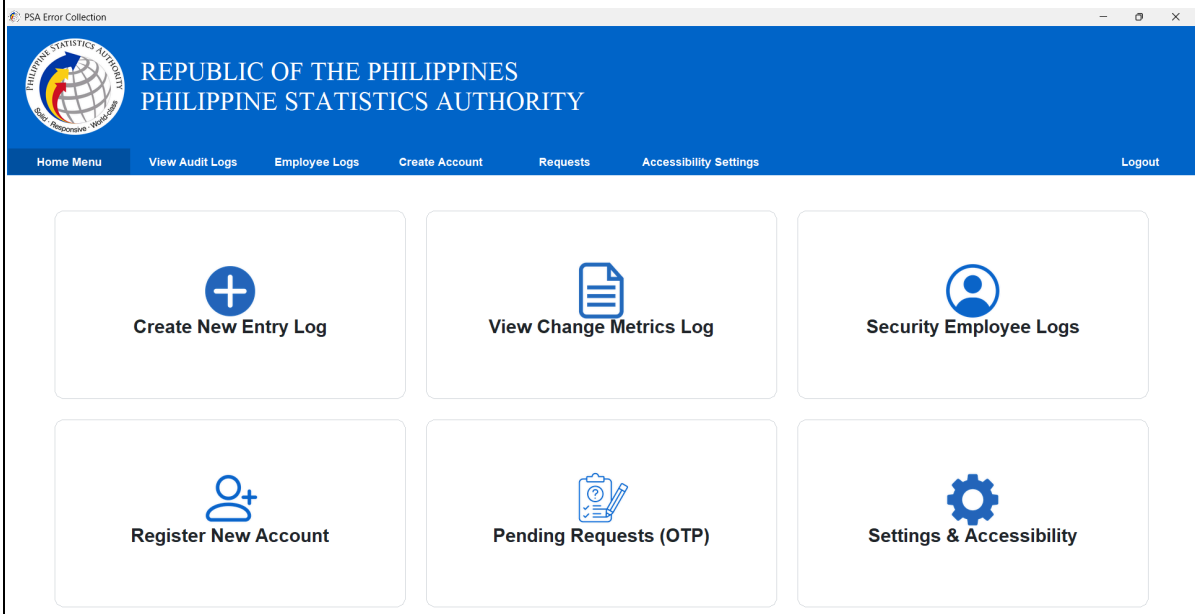


Figure 17: Admin-end Main Interface

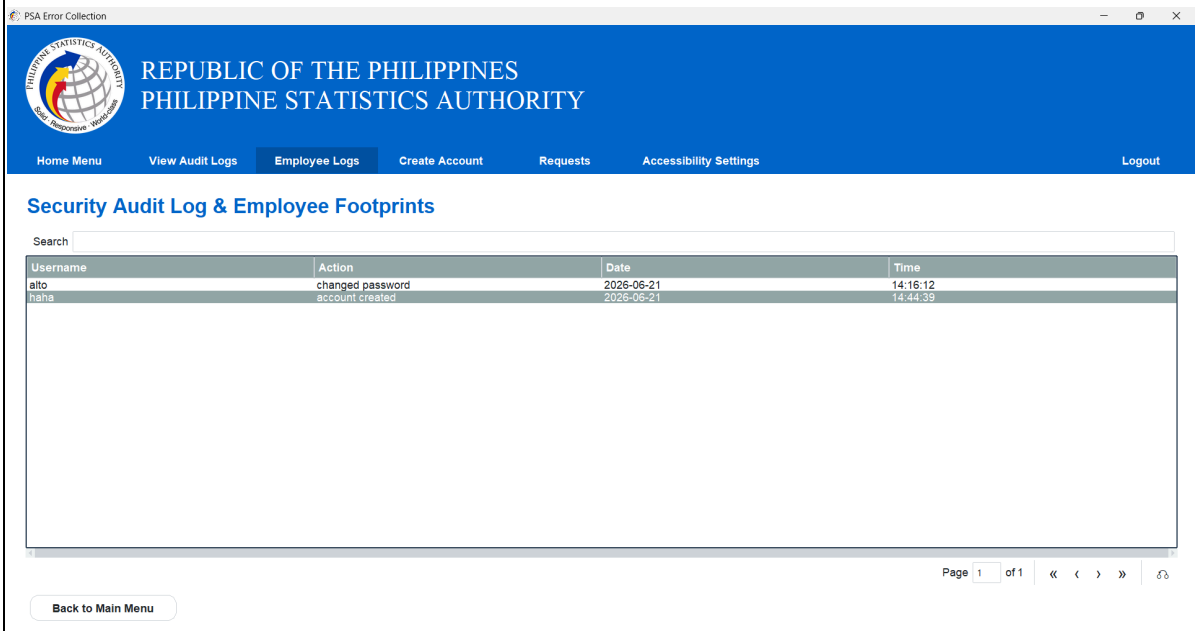


Figure 18: Employee Activity Log Interface

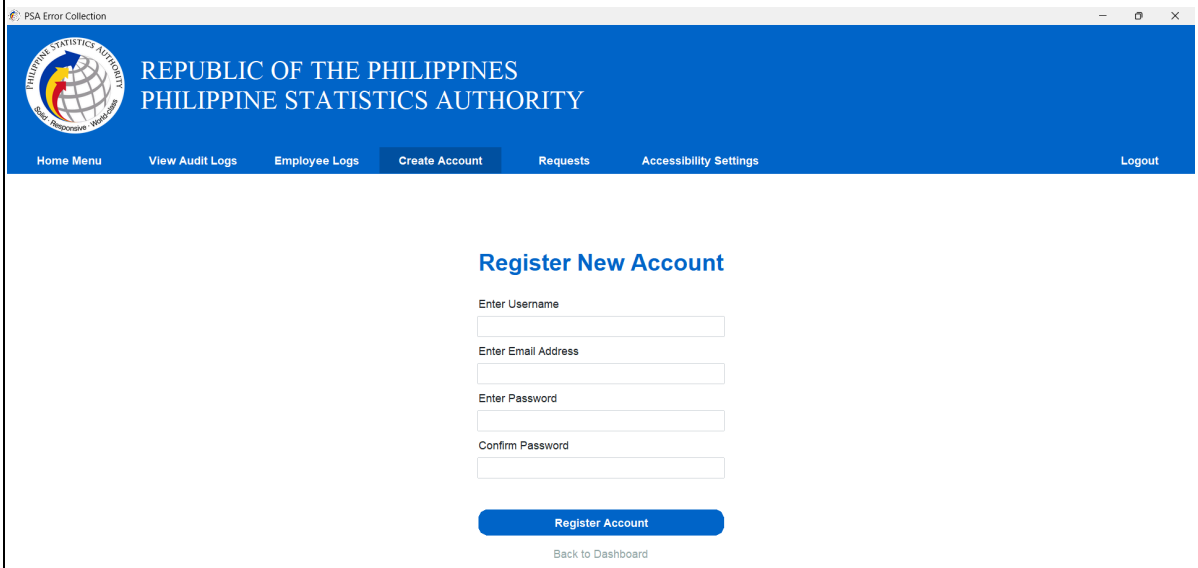


Figure 19: Create Account Interface

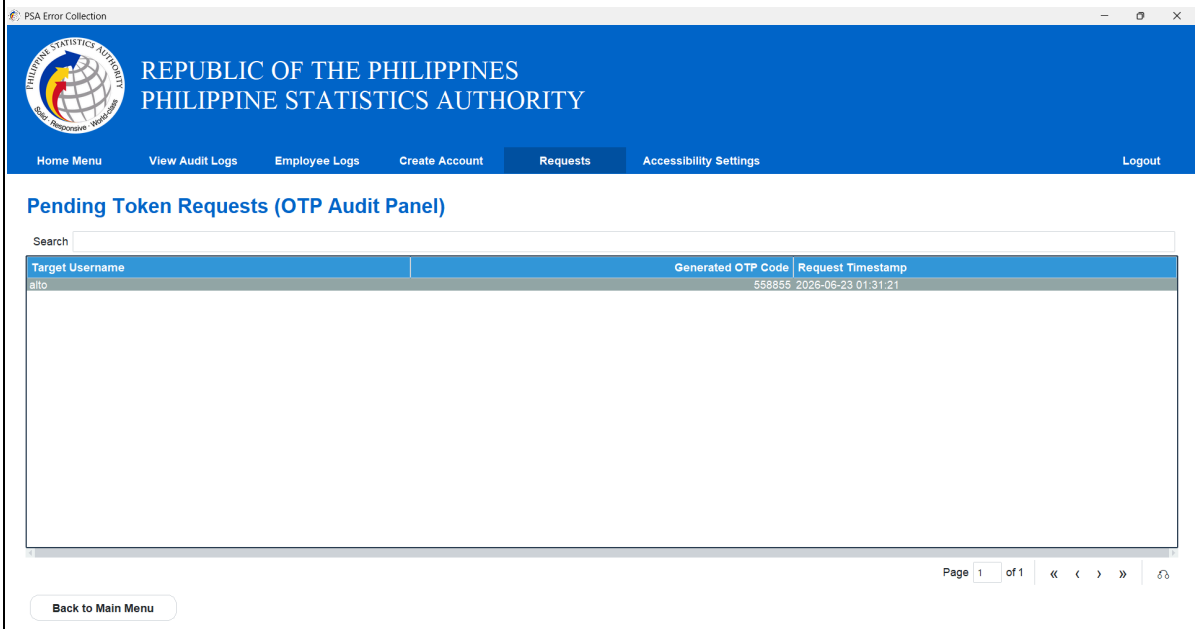


Figure 20: OTP Token Request Interface

SQL Queries

```sql
-- EASY MYSQL QUERIES

-- pending reports
SELECT *
FROM Discrepancy_Report
WHERE status = 'PENDING';

-- person who are born in 2000 millennium
SELECT *
FROM Person_data
WHERE birth_date >= 2000-01-01;

-- employees with M as first letter of first bame
SELECT *
FROM employees
WHERE emp_first_name LIKE 'M%';

-- Discrepancies with no explanation provided
SELECT *
FROM discrepancy_entrjes
WHERE explanation IS NULL;

-- Insert a new row
INSERT INTO Person_data
VALUES ( 'P-12345', 'Myrtle Isabella', 'Alcantara', 'Daitol', '2011-11-28');

-- Delete a row
DELETE FROM Person_data
WHERE person_id = 'P-12345';
```

Figure 21: Easy SQL Queries

```sql
WITH FormattedPersons AS (
    SELECT
        person_id,
        CONCAT_WS(' ', first_name, middle_name, last_name) AS full_name,
        last_name,
        first_name
    FROM person_data
)
SELECT
    rn.registry_number,  -- Now pulling from the REGISTRY_NUMBER table alias 'rn'
    c.full_name AS child_name,
    m.full_name AS mother_name,
    f.full_name AS father_name,
    CASE
        WHEN bc.birth_order = 1 THEN '1st'
        WHEN bc.birth_order = 2 THEN '2nd'
        WHEN bc.birth_order = 3 THEN '3rd'
        ELSE CONCAT(bc.birth_order, 'th')
    END AS birth_order
FROM birth_certificate bc
-- JOINING THE SEPARATE REGISTRY TABLE
-- Note: 'id' is a placeholder; replace with the actual matching column name
JOIN registry_number rn ON bc.child_id = rn.child_id
JOIN FormattedPersons c ON bc.child_id = c.person_id
JOIN FormattedPersons m ON bc.mother_id = m.person_id
JOIN FormattedPersons f ON bc.father_id = f.person_id
ORDER BY c.last_name ASC, c.first_name ASC;
```

Figure 22.1: Average name appending query

```sql
-- report status with assigned emp name
SELECT
    report.report_id,
    report.status,
    report.employee_id as assigned_employee_id,
    d.last_name as employee_lastname,
    d.first_name,
    d.middle_name
FROM
    workaround.discrepancy_report report
        INNER JOIN
    workaround.employee_data emp_id
        ON report.employee_id = emp_id.employee_id
        INNER JOIN
    workaround.person_data d
        ON emp_id.person_id = d.person_id
WHERE
    report.status LIKE '%PENDING%'
;
```

```sql
-- birth cert child names
SELECT
    birth.registry_number,
    birth.child_id,
    CONCAT(c.last_name,
            ', ',
            c.first_name,
            IFNULL(CONCAT(' ', LEFT(c.middle_name, 1), '.'),
                    '')) AS child_name,
    c.birth_date,
    birth.birth_order,
    -- TIMESTAMPDIFF(YEAR, d.birth_date, CURDATE()) AS age,
    CONCAT(m.last_name,
            ', ',
            m.first_name,
            IFNULL(CONCAT(' ', LEFT(m.middle_name, 1), '.'),
                    '')) AS mother_name,
    CONCAT(f.last_name,
            ', ',
            f.first_name,
            IFNULL(CONCAT(' ', LEFT(f.middle_name, 1), '.'),
                    '')) AS father_name

FROM
    workaround.birth_certificate birth
        INNER JOIN
    workaround.person_data c ON birth.child_id = c.person_id
        INNER JOIN
    workaround.person_data m ON birth.mother_id = m.person_id
        INNER JOIN
    workaround.person_data f ON birth.father_id = f.person_id
/*
WHERE
    d.birth_date <= '2000'
    or
    birth.birth_order = '2
*/
;
```

Figure 22.2: Average name appending query

```sql
-- Create and select the database
CREATE DATABASE IF NOT EXISTS Workaround;
USE Workaround;

-- (Optional) Create the workaround_entry table if it doesn't exist yet
CREATE TABLE IF NOT EXISTS workaround_entry (
    workaround_id VARCHAR(50) PRIMARY KEY,
    report_id VARCHAR(50),
    explanation VARCHAR(255),
    error_field VARCHAR(100),
    original_value VARCHAR(100),
    revised_value VARCHAR(100)
);

-- Clear the table if you want a fresh start
-- TRUNCATE TABLE workaround_entry;

-- ------------------------------------------------------------------------
-- 1. INSERT BIRTH CERTIFICATE ERRORS
-- ------------------------------------------------------------------------
INSERT INTO workaround_entry (workaround_id, report_id, explanation, error_field, original_value, revised_value)
SELECT
    CONCAT('E-B-', LPAD(ROW_NUMBER() OVER(ORDER BY bc.registry_number), 4, '0')),
    CONCAT('BC-26-', LPAD(MONTH(p.birth_date), 2, '0'), '-', LPAD(ROW_NUMBER() OVER(ORDER BY bc.registry_number), 5, '0')),
    'incorrect date of birth',
    'birth_order',
    CAST(bc.birth_order AS CHAR),
    CAST(CASE
        WHEN bc.birth_order = 1 THEN 2
        WHEN bc.birth_order = 2 THEN 1
        WHEN bc.birth_order = 3 THEN 2
    END AS CHAR)
FROM birth_certificate bc
JOIN person_data p ON bc.child_id = p.person_id
WHERE bc.birth_order IN (1, 2, 3);
```

```sql
-- ------------------------------------------------------------------------
-- 2. INSERT MARRIAGE CERTIFICATE ERRORS
-- ------------------------------------------------------------------------
INSERT INTO workaround_entry (workaround_id, report_id, explanation, error_field, original_value, revised_value)
SELECT
    CONCAT('E-M-', LPAD(ROW_NUMBER() OVER(ORDER BY registry_number), 4, '0')) AS workaround_id,
    CONCAT('MC-26-', LPAD(ROW_NUMBER() OVER(ORDER BY registry_number), 5, '0')) AS report_id,
    'Age discrepancy based on birth date' AS explanation,
    error_field,
    CAST(reported_age AS CHAR) AS original_value,
    CAST(actual_age AS CHAR) AS revised_value
FROM (
    -- Subquery: Applicant Age Check
    SELECT
        mc.registry_number,
        'applicant_age' AS error_field,
        mc.applicant_age AS reported_age,
        TIMESTAMPDIFF(YEAR, p.birth_date, CURDATE()) AS actual_age
    FROM marriage_certificate mc
    JOIN person_data p ON mc.applicant_id = p.person_id

    UNION ALL

    -- Subquery: Spouse Age Check
    SELECT
        mc.registry_number,
        'spouse_age' AS error_field,
        mc.spouse_age AS reported_age,
        TIMESTAMPDIFF(YEAR, p.birth_date, CURDATE()) AS actual_age
    FROM marriage_certificate mc
    JOIN person_data p ON mc.spouse_id = p.person_id
) AS Combined
WHERE reported_age <> actual_age;
```

```sql
-- ------------------------------------------------------------------------
-- 3. INSERT DEATH CERTIFICATE ERRORS
-- ------------------------------------------------------------------------
INSERT INTO workaround_entry (workaround_id, report_id, explanation, error_field, original_value, revised_value)
SELECT
    CONCAT('E-D-', LPAD(ROW_NUMBER() OVER(ORDER BY dc.registry_number), 4, '0')) AS workaround_id,
    CONCAT('DC-26-', LPAD(ROW_NUMBER() OVER(ORDER BY dc.registry_number), 5, '0')) AS report_id,
    'Certification date is after registration date' AS explanation,
    'registration_date / certification_date' AS error_field,
    CONCAT('Reg: ', CAST(dc.registration_date AS CHAR), ' | Cert: ', CAST(dc.certification_date AS CHAR)) AS original_value,
    CONCAT('Reg: ', CAST(dc.certification_date AS CHAR), ' | Cert: ', CAST(dc.registration_date AS CHAR)) AS revised_value
FROM death_certificate dc
WHERE dc.certification_date > dc.registration_date;


-- ------------------------------------------------------------------------
-- 4. VERIFY RESULTS
-- ------------------------------------------------------------------------
SELECT * FROM workaround_entry ORDER BY workaround_id;
```

Figure 23: DIfficult name appending query

System Specifications

This system is developed from Python Tkinter and modeled UI using ttk.bootstrap for themes, custom assets and buttons. Among the used modules also include the following:

re	Regular expressions for data validation
os	Operating system functions (file handling)
csv	CSV file reading/writing for account management
json	JSON file handling for audit logs
random	Generate random OTP codes for password reset
mysql.connector	MySQL database connection
MySQL Connector Error	MySQL error handling
ttk bootstrap	Modern themed UI widgets
	UI constants and styling
ttkbootstrap.widgets.tableview (Tableview)	Table display widget
tkinter. messagebox, StringVar	Dialog boxes and string variables

from datetime import datetime	Date/time tracking for logs
types	System-level Windows API calls for DPI scaling
PIL import Image, ImageTk	Image loading and display
traceback	Detailed error stack trace printing
sys	System module (used for PInstaller and resource resolution)

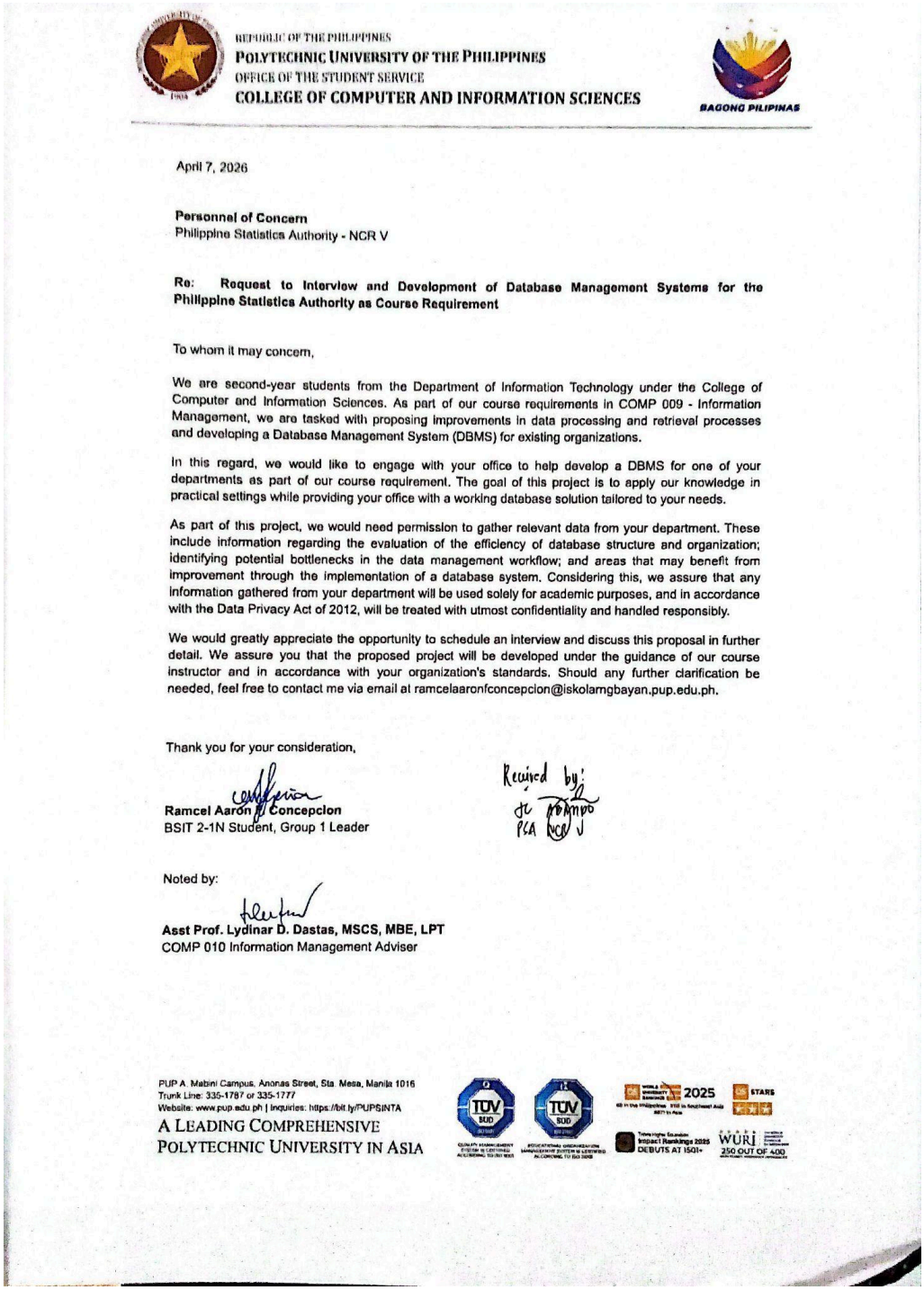
With that mentioned, it should be noted that the PSA Error Collection System is a lightweight desktop app that is compatible for most modern and semi-legacy units. To be more exact, the devices minimum requirements to run are as follows:

System Compatibility:
- Windows
- MacOS
- Linux

CPU Specifications:
- Intel Core i3 or
- AMD Ryzen 3
- 4GB RAM

Prerequisites
- Tkinter on VS Code
- Pillow Install
- mysql-connector-python
- MySQL Server (v5.7 or v8.0+)

REFERENCES AND DOCUMENTATION



REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE STUDENT SERVICE
COLLEGE OF COMPUTER AND INFORMATION SCIENCES

BAGONG PILIPINAS

April 7, 2026

Personnel of Concern
Philippine Statistics Authority - NCR V

Re: Request to Interview and Development of Database Management Systems for the Philippine Statistics Authority as Course Requirement

To whom it may concern,

We are second-year students from the Department of Information Technology under the College of Computer and Information Sciences. As part of our course requirements in COMP 009 - Information Management, we are tasked with proposing improvements in data processing and retrieval processes and developing a Database Management System (DBMS) for existing organizations.

In this regard, we would like to engage with your office to help develop a DBMS for one of your departments as part of our course requirement. The goal of this project is to apply our knowledge in practical settings while providing your office with a working database solution tailored to your needs.

As part of this project, we would need permission to gather relevant data from your department. These include information regarding the evaluation of the efficiency of database structure and organization; identifying potential bottlenecks in the data management workflow; and areas that may benefit from improvement through the implementation of a database system. Considering this, we assure that any information gathered from your department will be used solely for academic purposes, and in accordance with the Data Privacy Act of 2012, will be treated with utmost confidentiality and handled responsibly.

We would greatly appreciate the opportunity to schedule an interview and discuss this proposal in further detail. We assure you that the proposed project will be developed under the guidance of our course instructor and in accordance with your organization's standards. Should any further clarification be needed, feel free to contact me via email at ramcelaaronfconcepcion@iskolarngbayan.pup.edu.ph.

Thank you for your consideration,

Ramcel Aaron F. Concepcion
BSIT 2-1N Student, Group 1 Leader

Received by:
PSA NCR V

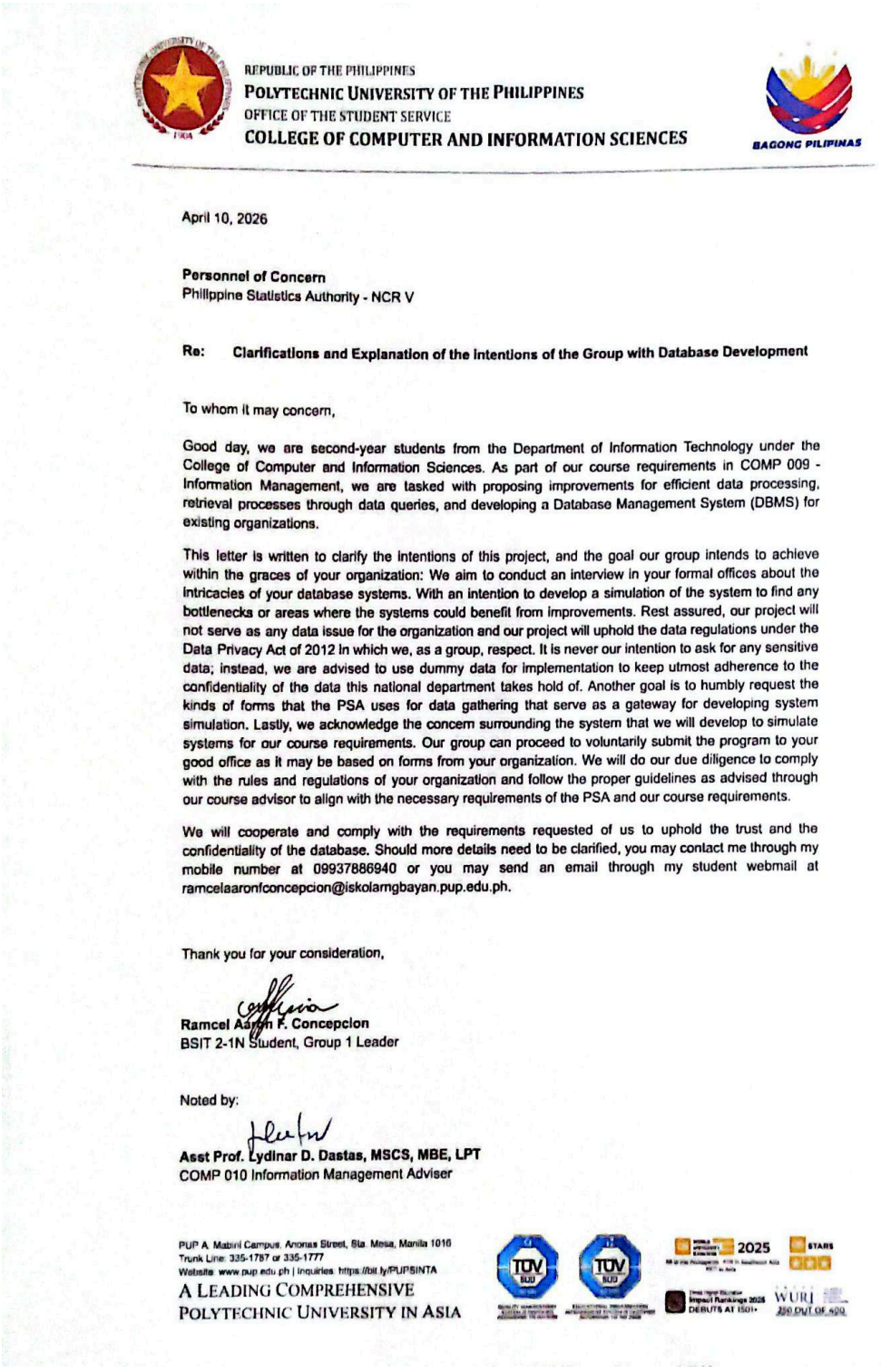
Noted by:

Asst Prof. Lydinar D. Dastas, MSCS, MBE, LPT
COMP 010 Information Management Adviser

PUP A. Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
Trunk Line: 335-1787 or 335-1777
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DEBUTS AT 1501+
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Letter 1: Initial Letter Request to the Philippine Statistics Authority



REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE STUDENT SERVICE
COLLEGE OF COMPUTER AND INFORMATION SCIENCES

BAGONG PILIPINAS

April 10, 2026

Personnel of Concern
Philippine Statistics Authority - NCR V

Re: Clarifications and Explanation of the Intentions of the Group with Database Development

To whom it may concern,

Good day, we are second-year students from the Department of Information Technology under the College of Computer and Information Sciences. As part of our course requirements in COMP 009 - Information Management, we are tasked with proposing improvements for efficient data processing, retrieval processes through data queries, and developing a Database Management System (DBMS) for existing organizations.

This letter is written to clarify the intentions of this project, and the goal our group intends to achieve within the graces of your organization: We aim to conduct an interview in your formal offices about the intricacies of your database systems. With an intention to develop a simulation of the system to find any bottlenecks or areas where the systems could benefit from improvements. Rest assured, our project will not serve as any data issue for the organization and our project will uphold the data regulations under the Data Privacy Act of 2012 in which we, as a group, respect. It is never our intention to ask for any sensitive data; instead, we are advised to use dummy data for implementation to keep utmost adherence to the confidentiality of the data this national department takes hold of. Another goal is to humbly request the kinds of forms that the PSA uses for data gathering that serve as a gateway for developing system simulation. Lastly, we acknowledge the concern surrounding the system that we will develop to simulate systems for our course requirements. Our group can proceed to voluntarily submit the program to your good office as it may be based on forms from your organization. We will do our due diligence to comply with the rules and regulations of your organization and follow the proper guidelines as advised through our course advisor to align with the necessary requirements of the PSA and our course requirements.

We will cooperate and comply with the requirements requested of us to uphold the trust and the confidentiality of the database. Should more details need to be clarified, you may contact me through my mobile number at 09937886940 or you may send an email through my student webmail at ramcelaaronfconcepcion@iskolarngbayan.pup.edu.ph.

Thank you for your consideration,

Ramcel Aaron F. Concepcion
BSIT 2-1N Student, Group 1 Leader

Noted by:

Asst Prof. Lydinar D. Dastas, MSCS, MBE, LPT
COMP 010 Information Management Adviser

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Letter 2: Clarification and Explanation of Intent to Develop Database

REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE STUDENT SERVICE
COLLEGE OF COMPUTER AND INFORMATION SCIENCES

BAGONG PILIPINAS

April 28, 2026

Personnel of Concern
Philippine Statistics Authority - NCR V

Re: Permission Request to Develop System for the PSA and Clarification From Previous Letters

To whom it may concern,

Good day, we are second-year students from the Department of Information Technology under the College of Computer and Information Sciences. As part of our course requirements in COMP 009 - Information Management, we are tasked with proposing improvements for efficient data processing, retrieval processes through data queries, and developing a Database Management System (DBMS) for existing organizations.

This letter is written to ask for your permission to allow us to improve upon your company's "workaround" system for erroneous data collection and correction. Our group's goal is to improve upon the current system by way of simulation and analysis to find potential improvement points. Rest assured, our project will not serve as any data issue for the organization and our project will uphold the data regulations under the Data Privacy Act of 2012 in which we, as a group, respect. It is never our intention to ask for any sensitive data; instead, we are advised to use dummy data for implementation to keep utmost adherence to the confidentiality of the data this national department takes hold of. Another goal is to once again ask for permission to work with your good office for our course requirements. Lastly, we acknowledge the concern surrounding the system that we will improve upon and what happens to the created simulation systems for our course requirements. Our group will voluntarily submit the program to your good office as it may be based on forms or models from your organization. We will do our due diligence to comply with the rules and regulations of your organization and follow the proper guidelines as advised through our course advisor to align with the necessary requirements of the PSA and our course requirements.

We will cooperate and comply with the requirements requested of us to uphold the trust and the confidentiality of the database. Should more details need to be clarified, you may contact me through my mobile number at 09937886940 or you may send an email through my student webmail at ramcelaaronfconcepcion@iskolarngbayan.pup.edu.ph.

Thank you for your consideration,

Ramcel Aaron F. Concepcion
BSIT 2-1N Student, Group 1 Leader

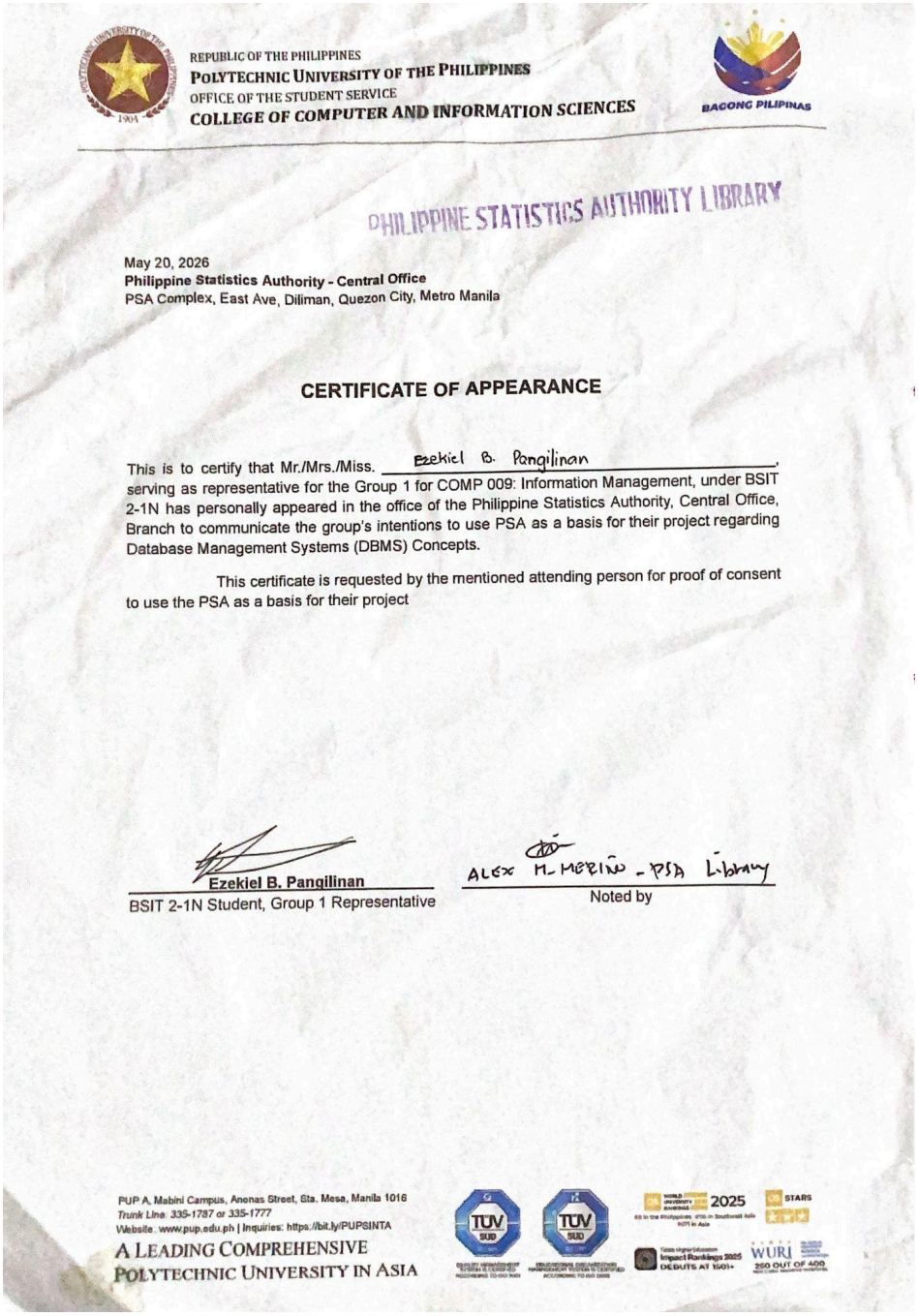
Noted by:

Asst Prof. Lydinar D. Dastas, MSCS, MBE, LPT
COMP 010 Information Management Adviser

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Letter 3: Permission Request to Implement System Improvements



REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE STUDENT SERVICE
COLLEGE OF COMPUTER AND INFORMATION SCIENCES

BAGONG PILIPINAS

PHILIPPINE STATISTICS AUTHORITY LIBRARY

May 20, 2026
Philippine Statistics Authority - Central Office
PSA Complex, East Ave, Diliman, Quezon City, Metro Manila

CERTIFICATE OF APPEARANCE

This is to certify that Mr./Mrs./Miss. Ezekiel B. Pangilinan, serving as representative for the Group 1 for COMP 009: Information Management, under BSIT 2-1N has personally appeared in the office of the Philippine Statistics Authority, Central Office, Branch to communicate the group's intentions to use PSA as a basis for their project regarding Database Management Systems (DBMS) Concepts.

This certificate is requested by the mentioned attending person for proof of consent to use the PSA as a basis for their project

Ezekiel B. Pangilinan
BSIT 2-1N Student, Group 1 Representative

ALEX M. MERIÑO - PSA Library
Noted by

PUP A, Mabini Campus, Anonas Street, Sta. Mesa, Manila 1016
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Website: www.pup.edu.ph | Inquiries: https://bit.ly/PUPSINTA
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Letter 4: Certificate of Appearance for Approval

Bibliography

The group would like to acknowledge the sources used that were instrumental in the development and completion of this project. The group would like to cite the following sources for giving insight into the substance needed for the system:

- PSAHelpline.Ph. (2024, November 26). *Understanding errors in your PSA birth certificate and what to do next*. https://psahelpline.ph/blogs/understanding-errors-in-your-psa-birth-certificate-and-what-to-do-next

- *Wrong spelling | Philippine Statistics Authority*. (n.d.). https://psa.gov.ph/civilregistration/problems-and-solutions/wrong-spelling

- *PSA launches APCAS for digital, seamless processing of petition for corrections in civil registry documents Philippine Statistics Authority*. (2026, May 22). https://psa.gov.ph/content/psa-launches-apcas-digital-seamless-processing-petition-corrections-civil-registry\
- PSAHelpline.Ph. (2024, November 26). PSA Birth Certificate: Correcting interchanged middle and last names. https://psahelpline.ph/blogs/psa-birth-certificate-correcting-interchanged-middle-and-last-names